

Statistical Properties of the OLS Coefficient Estimators

We have derived the OLS (Ordinary Least Squares) estimators of the regression coefficients in the simple linear regression model given by the **population regression equation**, or PRE

$$Y_i = \beta_0 + \beta_1 X_i + u_i \quad (i = 1, \dots, n) \quad (1)$$

where u_i is an iid random error term. The **OLS sample regression equation** (SRE) corresponding to PRE (1) is

$$Y_i = \hat{\beta}_0 + \hat{\beta}_1 X_i + \hat{u}_i \quad (i = 1, \dots, n) \quad (2)$$

where $\hat{\beta}_0$ and $\hat{\beta}_1$ are the **OLS coefficient estimators** given by the formulas

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2} \quad (3)$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \quad (4)$$

$$x_i \equiv X_i - \bar{X}, \quad y_i \equiv Y_i - \bar{Y}, \quad \bar{X} = \sum_i X_i / n \text{ and } \bar{Y} = \sum_i Y_i / n$$

Why Use the OLS Coefficient Estimators?

The reason we use these OLS coefficient estimators is that, under assumptions A1-A8 of the classical linear regression model, they have several desirable statistical properties. We are going to examine these desirable statistical properties of the OLS coefficient estimators primarily in terms of the OLS slope coefficient estimator; the same properties apply to the intercept coefficient estimator.

➤ **PROPERTY 1: Linearity of $\hat{\beta}_1$**

The OLS coefficient estimator $\hat{\beta}_1$ can be written as a **linear function of the sample values of Y, the Y_i ($i = 1, \dots, n$)** .

Proof: Starts with formula (3) for $\hat{\beta}_1$:

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_i x_i y_i}{\sum_i x_i^2} \\ &= \frac{\sum_i x_i (Y_i - \bar{Y})}{\sum_i x_i^2} \\ &= \frac{\sum_i x_i Y_i}{\sum_i x_i^2} - \frac{\bar{Y} \sum_i x_i}{\sum_i x_i^2} \\ &= \frac{\sum_i x_i Y_i}{\sum_i x_i^2}\end{aligned}$$

because $\sum_i x_i = 0$.

- Defining the observation weights $k_i = x_i / \sum_i x_i^2$ for $(i = 1, \dots, n)$ we can re-write the last expression above for $\hat{\beta}_1$ as:

$$\hat{\beta}_1 = \sum_i k_i Y_i \quad \text{where } k_i \equiv \frac{x_i}{\sum_i x_i^2} \quad (i = 1, \dots, n) \quad \dots \text{(P1)}$$

- Note that the formula (3) and the definition of the weights k_i imply that $\hat{\beta}_1$ is also a linear function of the y_i 's such that

$$\hat{\beta}_1 = \sum_i k_i y_i.$$

- **Result:** The OLS slope coefficient estimator $\hat{\beta}_1$ is a **linear function of the sample values Y_i or y_i** ($i = 1, \dots, n$) where the coefficient of Y_i or y_i is k_i .

□ Properties of the Weights k_i

In order to establish the remaining properties of $\hat{\beta}_1$, it is necessary to know the arithmetic properties of the weights k_i .

[K1] $\sum_i k_i = 0$, i.e., the weights k_i sum to zero.

$$\sum_i k_i = \sum_i \frac{x_i}{\sum_i x_i^2} = \frac{1}{\sum_i x_i^2} \sum_i x_i = 0, \quad \text{because } \sum_i x_i = 0.$$

$$[\text{K2}] \quad \sum_i k_i^2 = \frac{1}{\sum_i x_i^2}.$$

$$\sum_i k_i^2 = \sum_i \left(\frac{x_i}{\sum_i x_i^2} \right)^2 = \sum_i \frac{x_i^2}{\left(\sum_i x_i^2 \right)^2} = \frac{\left(\sum_i x_i^2 \right)}{\left(\sum_i x_i^2 \right)^2} = \frac{1}{\sum_i x_i^2}.$$

$$[\text{K3}] \quad \sum_i k_i x_i = \sum_i k_i X_i .$$

$$\begin{aligned} \sum_i k_i x_i &= \sum_i k_i (X_i - \bar{X}) \\ &= \sum_i k_i X_i - \bar{X} \sum_i k_i \\ &= \sum_i k_i X_i \end{aligned}$$

since $\sum_i k_i = 0$ by [K1] above.

$$[\text{K4}] \quad \sum_i k_i x_i = 1 .$$

$$\sum_i k_i x_i = \sum_i \left(\frac{x_i}{\sum_i x_i^2} \right) x_i = \sum_i \frac{x_i^2}{\left(\sum_i x_i^2 \right)} = \frac{\left(\sum_i x_i^2 \right)}{\left(\sum_i x_i^2 \right)} = 1 .$$

Implication: $\sum_i k_i X_i = 1 .$

➤ **PROPERTY 2: Unbiasedness of $\hat{\beta}_1$ and $\hat{\beta}_0$.**

The OLS coefficient estimator $\hat{\beta}_1$ is unbiased, meaning that $E(\hat{\beta}_1) = \beta_1$.

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- **Definition of unbiasedness:** The coefficient estimator $\hat{\beta}_1$ is unbiased if and only if $E(\hat{\beta}_1) = \beta_1$; i.e., its mean or expectation is equal to the true coefficient β_1 .

- **Proof of unbiasedness of $\hat{\beta}_1$:** Start with the formula $\hat{\beta}_1 = \sum_i k_i Y_i$.

1. Since assumption A1 states that the PRE is $Y_i = \beta_0 + \beta_1 X_i + u_i$,

$$\hat{\beta}_1 = \sum_i k_i Y_i$$

$$= \sum_i k_i (\beta_0 + \beta_1 X_i + u_i)$$

since $Y_i = \beta_0 + \beta_1 X_i + u_i$ by A1

$$= \beta_0 \sum_i k_i + \beta_1 \sum_i k_i X_i + \sum_i k_i u_i$$

$$= \beta_1 + \sum_i k_i u_i,$$

since $\sum_i k_i = 0$ and $\sum_i k_i X_i = 1$.

2. Now take expectations of the above expression for $\hat{\beta}_1$, conditional on the sample values $\{X_i: i = 1, \dots, n\}$ of the regressor X . Conditioning on the sample values of the regressor X means that the k_i are treated as nonrandom, since the k_i are functions only of the X_i .

$$\begin{aligned} E(\hat{\beta}_1) &= E(\beta_1) + E[\sum_i k_i u_i] \\ &= \beta_1 + \sum_i k_i E(u_i | X_i) \quad \text{since } \beta_1 \text{ is a constant and the } k_i \text{ are nonrandom} \\ &= \beta_1 + \sum_i k_i 0 \quad \text{since } E(u_i | X_i) = 0 \text{ by assumption A2} \\ &= \beta_1. \end{aligned}$$

- **Result:** The OLS slope coefficient estimator $\hat{\beta}_1$ is an *unbiased* estimator of the slope coefficient β_1 : that is,

$$E(\hat{\beta}_1) = \beta_1. \quad \dots \text{ (P2)}$$

- **Proof of unbiasedness of $\hat{\beta}_0$:** Start with the formula $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$.

1. Average the PRE $Y_i = \beta_0 + \beta_1 X_i + u_i$ across i :

$$\sum_{i=1}^n Y_i = n \beta_0 + \beta_1 \sum_{i=1}^n X_i + \sum_{i=1}^n u_i \quad (\text{sum the PRE over the } N \text{ observations})$$

$$\frac{\sum_{i=1}^n Y_i}{n} = \frac{n \beta_0}{n} + \beta_1 \frac{\sum_{i=1}^n X_i}{n} + \frac{\sum_{i=1}^n u_i}{n} \quad (\text{divide by } n)$$

$$\bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{u} \quad \text{where } \bar{Y} = \sum_i Y_i / n, \bar{X} = \sum_i X_i / n, \text{ and } \bar{u} = \sum_i u_i / n.$$

2. Substitute the above expression for $\bar{Y}$ into the formula $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$= \beta_0 + \beta_1 \bar{X} + \bar{u} - \hat{\beta}_1 \bar{X} \quad \text{since } \bar{Y} = \beta_0 + \beta_1 \bar{X} + \bar{u}$$

$$= \beta_0 + (\beta_1 - \hat{\beta}_1) \bar{X} + \bar{u}.$$

3. Now take the expectation of $\hat{\beta}_0$ conditional on the sample values $\{X_i: i = 1, \dots, n\}$ of the regressor X . Conditioning on the X_i means that $\bar{X}$ is treated as nonrandom in taking expectations, since X is a function only of the X_i .

$$\begin{aligned} E(\hat{\beta}_0) &= E(\beta_0) + E[(\beta_1 - \hat{\beta}_1)\bar{X}] + E(\bar{u}) \\ &= \beta_0 + \bar{X} E(\beta_1 - \hat{\beta}_1) + E(\bar{u}) \quad \text{since } \beta_0 \text{ is a constant} \\ &= \beta_0 + \bar{X} E(\beta_1 - \hat{\beta}_1) \quad \text{since } E(\bar{u}) = 0 \text{ by assumptions A2 and A5} \\ &= \beta_0 + \bar{X} [E(\beta_1) - E(\hat{\beta}_1)] \\ &= \beta_0 + \bar{X} (\beta_1 - \beta_1) \quad \text{since } E(\beta_1) = \beta_1 \text{ and } E(\hat{\beta}_1) = \beta_1 \\ &= \beta_0 \end{aligned}$$

- **Result:** The OLS intercept coefficient estimator $\hat{\beta}_0$ is an *unbiased* estimator of the intercept coefficient β_0 : that is,

$$E(\hat{\beta}_0) = \beta_0. \quad \dots \text{ (P2)}$$

➤ **PROPERTY 3:** Variance of $\hat{\beta}_1$ is minimum in the class of all linear and unbiased estimators.

- **Definition:** The variance of the OLS slope coefficient estimator $\hat{\beta}_1$ is defined as

$$\text{Var}(\hat{\beta}_1) \equiv E\left\{\left[\hat{\beta}_1 - E(\hat{\beta}_1)\right]^2\right\}.$$

- **Derivation of Expression for $\text{Var}(\hat{\beta}_1)$:**

1. Since $\hat{\beta}_1$ is an unbiased estimator of β_1 , $E(\hat{\beta}_1) = \beta_1$. The variance of $\hat{\beta}_1$ can therefore be written as

$$\text{Var}(\hat{\beta}_1) = E\left\{\left[\hat{\beta}_1 - \beta_1\right]^2\right\}.$$

2. From part (1) of the unbiasedness proofs above, the term $[\hat{\beta}_1 - \beta_1]$, which is called the **sampling error of $\hat{\beta}_1$** , is given by

$$[\hat{\beta}_1 - \beta_1] = \sum_i k_i u_i.$$

3. The square of the sampling error is therefore

$$[\hat{\beta}_1 - \beta_1]^2 = (\sum_i k_i u_i)^2$$

4. Since the square of a sum is equal to the sum of the squares plus twice the sum of the cross products,

$$\begin{aligned} [\hat{\beta}_1 - \beta_1]^2 &= (\sum_i k_i u_i)^2 \\ &= \sum_{i=1}^n k_i^2 u_i^2 + 2 \sum_{i < s} \sum_{s=2}^n k_i k_s u_i u_s. \end{aligned}$$

For example, if the summation involved only three terms, the square of the sum would be

$$\begin{aligned}\left(\sum_{i=1}^3 k_i u_i\right)^2 &= (k_1 u_1 + k_2 u_2 + k_3 u_3)^2 \\ &= k_1^2 u_1^2 + k_2^2 u_2^2 + k_3^2 u_3^2 + 2k_1 k_2 u_1 u_2 + 2k_1 k_3 u_1 u_3 + 2k_2 k_3 u_2 u_3.\end{aligned}$$

5. Now use assumptions A3 and A4 of the classical linear regression model (CLRM):

$$(A3) \quad \text{Var}(u_i | X_i) = E(u_i^2 | X_i) = \sigma^2 > 0 \quad \text{for all } (i = 1, \dots, n)$$

$$(A4) \quad \text{Cov}(u_i, u_s | X_i, X_s) = E(u_i u_s | X_i, X_s) = 0 \text{ for all } i \neq s.$$

6. We take expectations conditional on the sample values of the regressor X :

$$\begin{aligned}
 E\left[\left(\hat{\beta}_1 - \beta_1\right)^2\right] &= \sum_{i=1}^n k_i^2 E(u_i^2 | X_i) + 2 \sum_{i < s} \sum_{s=2}^n k_i k_s E(u_i u_s | X_i, X_s) \\
 &= \sum_{i=1}^n k_i^2 E(u_i^2 | X_i) \quad \text{since } E(u_i u_s | X_i, X_s) = 0 \text{ for } i \neq s \text{ by (A4)} \\
 &= \sum_{i=1}^n k_i^2 \sigma^2 \quad \text{since } E(u_i^2 | X_i) = \sigma^2 \quad \forall i \text{ by (A3)} \\
 &= \frac{\sigma^2}{\sum_i x_i^2} \quad \text{since } \sum_i k_i^2 = \frac{1}{\sum_i x_i^2} \text{ by (K2).}
 \end{aligned}$$

□ **Result:** The *variance* of the OLS slope coefficient estimator $\hat{\beta}_1$ is

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_i x_i^2} = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2} = \frac{\sigma^2}{\text{TSS}_X} \quad \text{where } \text{TSS}_X = \sum_i x_i^2. \quad \dots \text{ (P3)}$$

$$\hat{\beta}_1 = \sum k_i Y_i$$

where

$$k_i = \frac{X_i - \bar{X}}{\sum (X_i - \bar{X})^2} = \frac{x_i}{\sum x_i^2} \quad (\text{see Appendix 3A.2})$$

which shows that $\hat{\beta}_1$ is a weighted average of the Y 's, with k_i serving as the weights.

Let us define an alternative linear estimator of β_1 as follows:

$$\beta_1^* = \sum w_i Y_i$$

where w_i are also weights, not necessarily equal to k_i . Now

$$\begin{aligned} E(\beta_1^*) &= \sum w_i E(Y_i) \\ &= \sum w_i (\beta_0 + \beta_1 X_i) \\ &= \beta_0 \sum w_i + \beta_1 \sum w_i X_i \end{aligned}$$

Therefore, for β_1^* to be unbiased, we must have

$$\sum w_i = 1$$

and

$$\sum w_i X_i = \bar{X}$$

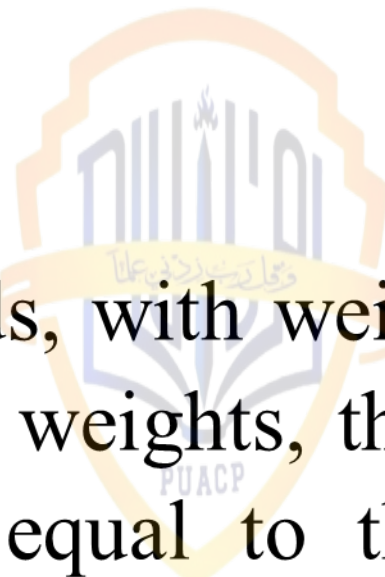
Also, we may write

$$\begin{aligned}\text{var}(\beta_1^*) &= \text{var} \sum w_i Y_i \\&= \sum w_i^2 \text{var} Y_i \quad [\text{Note: } \text{var} Y_i = \text{var} u_i = \sigma^2] \\&= \sigma^2 \sum w_i^2 \quad [\text{Note: } \text{cov}(Y_i, Y_j) = 0 (i \neq j)] \\&= \sigma^2 \sum \left(w_i - \frac{x_i}{\sum x_i^2} + \frac{x_i}{\sum x_i^2} \right)^2 \quad (\text{Note the mathematical trick}) \\&= \sigma^2 \sum \left(w_i - \frac{x_i}{\sum x_i^2} \right)^2 + \sigma^2 \frac{\sum x_i^2}{(\sum x_i^2)^2} + 2\sigma^2 \sum \left(w_i - \frac{x_i}{\sum x_i^2} \right) \left(\frac{x_i}{\sum x_i^2} \right) \\&= \sigma^2 \sum \left(w_i - \frac{x_i}{\sum x_i^2} \right)^2 + \sigma^2 \left(\frac{1}{\sum x_i^2} \right)\end{aligned}$$

because the last term in the next to the last step drops out. (Why?)

Since the last term in above equation is constant, the variance of (β_2^*) can be minimized only by manipulating the first term. If we let

$$\begin{aligned}w_i &= \frac{x_i}{\sum x_i^2} \\ \text{var}(\beta_1^*) &= \frac{\sigma^2}{\sum x_i^2} \\ &= \text{var}(\hat{\beta}_2)\end{aligned}$$



In words, with weights $w_i = k_i$, which are the least-squares weights, the variance of the linear estimator β_1^* is equal to the variance of the least-squares estimator $\widehat{\beta}_1$; otherwise $\text{var}(\beta_1^*) > \text{var}(\widehat{\beta}_1)$. To put it differently, if there is a minimum-variance linear unbiased estimator of β_1 , it must be the least squares estimator of β_1 .